

Automated Characterization Platform for Comprehensive Dynamic R_{dson} Assessment of GaN HEMTs From 50 K to 400 K

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Abstract—Gallium Nitride (GaN) HEMTs demonstrate superior performance at cryogenic temperatures, making them promising candidates for lunar missions where temperatures vary from 50 to 400 K. However, the current collapse effect increases dynamic on-state resistance (R_{dson}) during switching, leading to additional energy loss, especially in high-frequency applications. To conduct a comprehensive study with high efficiency, we developed an automated characterization platform for dynamic R_{dson} extraction under multi-stressor conditions. The design considerations include 1) programmable stressors such as temperatures, voltages, currents, etc., 2) electrothermal-mechanical compatibility with the cryocooler across a wide temperature range, and 3) software framework for full automation. Results for a commercial GaN HEMT show that dynamic R_{dson} varies non-monotonically with temperature increment, as it increases at 50 – 180 K and then decreases a 180 – 400 K. Dynamic R_{dson} follows this trend at both hard- and soft-switching, but is more sensitive to stressors at hard switching, while remaining nearly unchanged at soft switching, indicating that, under this testing plan that applies low stressing voltages, hot electron effect makes the major contribution to dynamic R_{dson} .

Keywords—GaN HEMT, dynamic characterization, cryogenic test, automated platform

I. INTRODUCTION

Power electronics for space applications have attracted significant attention in recent years, such as the solar-powered systems for lunar mission. These systems, however, face extreme environmental challenges on the lunar surface, where temperatures range from 400 K during the day to 50 K at night [1]. Gallium Nitride (GaN) high electron mobility transistors (HEMTs) offer significant advantages over silicon devices, including fast switching speed and low conduction loss [2]. As a promising candidate for space applications, GaN HEMTs possess great potential especially at cryogenic temperatures, in terms of, for example, generally lower ON-state resistance (R_{dson}) at low temperatures [3].

Nevertheless, GaN HEMTs suffer from dynamic ON-state resistance (D- R_{dson}) effect: At the onset that the device is turned on, its actual R_{dson} is notably higher than the static R_{dson} [4]. As shown in Fig. 1, the extra R_{dson} introduces more energy losses and things will be even worse when the switching frequency becomes higher [5]. Therefore, it is essential to characterize the behavior of D- R_{dson} at a wide temperature range, in such way

that a comprehensive database can be established to support the modeling and design of GaN-based converters for space applications.

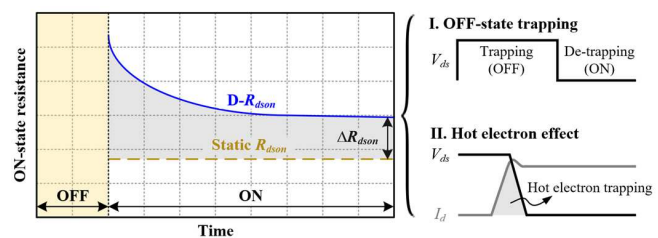


Fig. 1. Dynamic R_{dson} and its physical origins.

The major physical origins of D- R_{dson} includes two aspects: I. Electron trapping (capture) during the OFF state when the device is blocking a high DC bus voltage [6], and II. Hot electron effect that occurs during the switching transient when a large electric field and large current exist simultaneously in the drain region [7]. Thus, on top of temperature, D- R_{dson} can be impacted by multiple factors or stressors, including the amplitude of blocking voltage and conducted current, the stressing time (i.e., the duration of being biased by high voltage), as well as switching mode since the hot electron effect is more severe under hard-switching conditions. In addition, the switching speed is also impactful since it is related to switching loss and hot electron effect. As presented in TABLE I, existing research has performed comprehensive studies on D- R_{dson} [8]–[17]. Generally, it includes 1) designing circuit topology to enable flexible stressor configuration and reduce self-heating of the device under test (DUT) [8], 2) designing clamping circuit for fast and accurate switching transient measurement [10], 3) studying the physical mechanism and analyzing stressors' impact [11] and 4) modeling the dynamic behavior via equivalent circuit or analytical approaches [16].

However, the study of D- R_{dson} behavior under multi-stressor conditions, especially in a wide temperature range like 50 – 400 K, has not been fully explored yet. In addition, as reported in Ref. [5], discrepancies exist among the results from different research studies due to variations in test setups. Unlike the static characterization that can be performed using a curve tracer, the dynamic characterization largely depends on specifically developed testing platforms. Despite the JEDEC standard that

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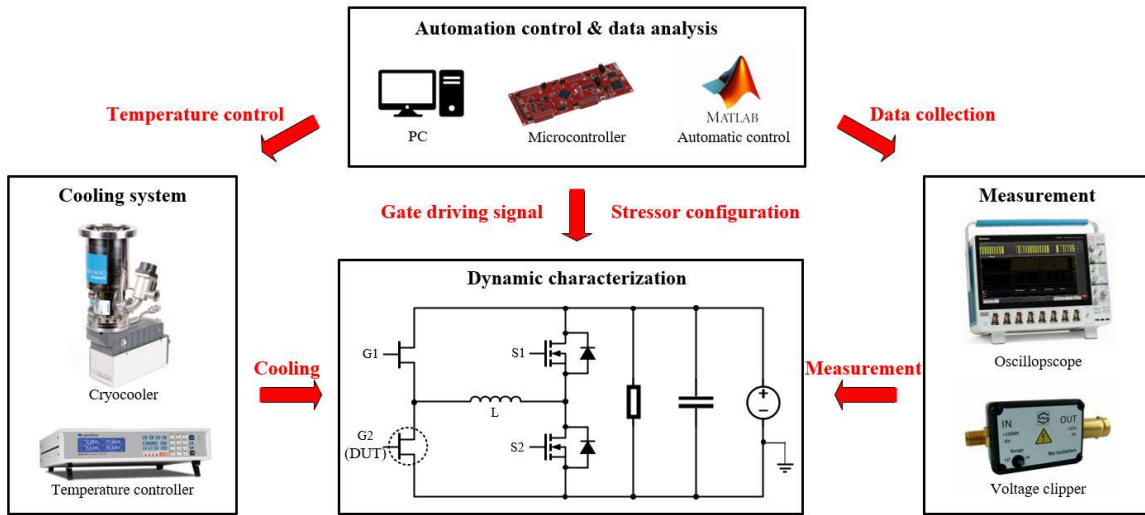


Fig. 2. Platform overview, including function modules of control, cooling, testing and measurement.

recommends double pulse test (DPT) for $D-R_{ds(on)}$ measurement [18], the difference in circuit parameters and PCB layout design still challenges the fair comparison among devices. On top of that, performing a comprehensive test accounting for as many stressors will significantly increase the testing workload, making it practically challenging for manual operation.

To address these issues, we propose to develop a fully automated platform for comprehensive $D-R_{ds(on)}$ measurement from 50 to 400 K. The platform automatically programs various stressors—temperatures, switching modes, voltages, currents, etc.—allowing comprehensive characterization of GaN devices with minimal human effort. In Section II, the platform overview is introduced, involving the manipulation of each stressor and assembly of functional modules at system level. In Section III, the design considerations of the testing circuit is discussed, in terms of both hardware and software. In Section IV, the platform is implemented to collect $D-R_{ds(on)}$ data, by which the impact of each stressor is analyzed. Section V concludes the work.

II. PLATFORM OVERVIEW

Fig. 2 shows the platform overview with functional module assembly. In this work, the investigated impacting factors / stressors are as follows:

- Temperature
- Switching mode

- Blocking voltage
- Drain current
- Stressing time

The configuration of each stressor is achieved by automatic control using the computer and MATLAB, which can be accessed with remote desktop. The details are given as follows.

(1) Temperature control

The cooling system comprises mainly a cryocooler and a temperature controller. The cryocooler cools down the DUT with a LHe_2 -compressor performing closed-loop helium cycling. A silicon diode sensor is implemented to measure the temperature, and once it approaches the target, the temperature controller will activate the heater inside the cryocooler to balance the thermal process, such that the final temperature is stabilized around the target. The temperature controller is managed by a developed MATLAB software that sends out commands via serial interface, adjusting temperature target, slew rate, heater power, etc. at each test round.

(2) Switching mode and stressing time control

The control of switching mode and stressing time is achieved by configuring the sequence and duration of turn-on and turn-off of S1, S2, G1, and G2 (DUT) in Fig. 2. MATLAB generates corresponding gate control timing, and then sends commands

TABLE I. SUMMARY OF STATE-OF-THE-ART STUDIES OF $D-R_{ds(on)}$.

	Impacting factors									Mechanism	Modeling	
	V_{DS}	I_D	t_{stress}	f_{sw}	Pulse no.	Sw. speed	Sw. mode	Temperature				
								High	Room			Cryogenic
Prior work												
Extensive Moderate Limited												
Method	Testing circuit with timing control					Gate driver	Hard- & soft sw.	Thermal chamber			TCAD simulation	Circuit or analytical models

through serial interface to the microcontroller (MCU) to conduct turn-on and -off of each transistor. The detailed timing design is presented in the next section.

(3) Blocking voltage control

A 600-V DC power supply is used to apply the bus voltage, which is controlled by the MATLAB command to accordingly adjust the output voltage. When performing the test, MATLAB leverages the pre-defined voltage output percentage and the DUT's rating to determine the actually required bus voltage.

(4) Drain current control

The current through the device is controlled by the charging voltage and duration of the load inductor in the same way as DPT [20]. During the test, MATLAB uses the pre-defined current output percentage, the bus voltage and inductance to determine the duration of turn-on and -off of each transistor to generate desired current. Details are given in the next section.

III. DESIGN CONSIDERATIONS FOR COMPREHENSIVE TEST

A. Circuit Design and timing control

According to JEDEC standard, the double pulse test (DPT) is used for $D-R_{dson}$ extraction [18]. The circuit diagram is shown in Fig. 3. G1 and G2 are GaN HEMTs, with G2 acting as the DUT for $D-R_{dson}$ extraction. Two additional SiC MOSFETs (S1 and S2) are used to adjust the switching modes. Each stressor (voltage, current, stressing time, etc.) is independently regulated by the computer controlling V_{DC} and switching timing of transistors.

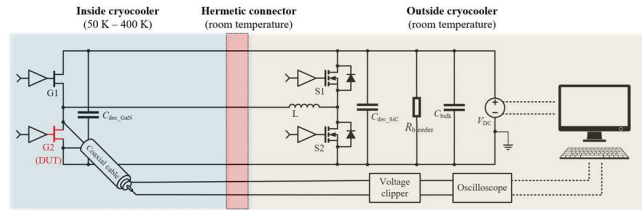


Fig. 3. Functional diagram of circuitry.

The stressors are adjusted by controlling the turn-on / -off of G1, G2, S1 and S2, which is implemented via ePWM function of the MCU. A detailed illustration is shown in Fig. 4. The scheme comprises three sections: de-trapping, trapping and stressing time. During the de-trapping section, only the DUT is turned on to avoid and unexpected bias affecting the results and to ensure the device is free of the trapping effect from the last test. Considering the time constants of de-trapping [16], [19], the duration of this section is set to be 10 seconds. As shown in Fig. 5, in hard-switching mode, after de-trapping, the operation is the same as the conventional DPT setup, where t_{charge} is determined by Eq. (1). In the trapping section, the DUT is biased by V_{DC} and I_L freewheels through G1 and S1, by changing the duration of which, different stressing time is applied in the test. On top of the extraction of $D-R_{dson}$, the hard-switching operation is also capable of measuring the static R_{dson} during the 1st pulse of G2, right after I_L starting increasing once S1 is turned on. The measured static R_{dson} is ensured to be free of trapping effect and hot electron effect because of the sufficiently long de-trapping time, which guarantees a fair result.

$$t_{charge} = \frac{LI_L}{V_{DC}} \quad (1)$$

In terms of soft-switching mode, however, additional pre-stressing time is required before charging the inductor, since the charging and trapping are coupled. As shown in Fig. 5, in the current charging stage, G1 and S2 are turned on to increase I_L , and simultaneously the DUT is biased by V_{DC} . Suppose the target stressing (trapping) time is t_{stress} , therefore, a pre-stressing time t_{pre} , which is determined in Eq. (2), is desired to have the DUT in the trapping section ahead of charging of I_L .

$$t_{pre} = t_{stress} - t_{charge} \quad (2)$$

By introducing t_{pre} , the stressing time and charging time can be decoupled, making it flexible to control the trapping process.

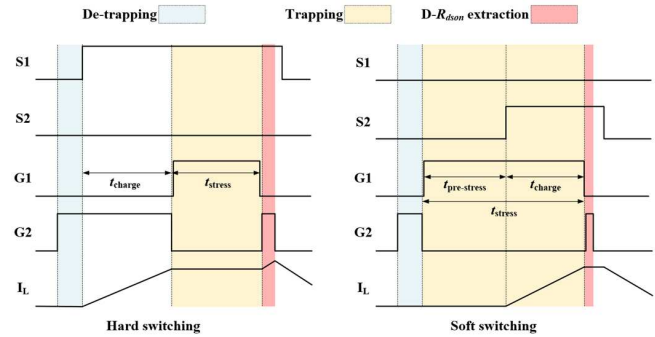


Fig. 4. Timing control scheme at hard- and soft-switching conditions.

The design methodology of DPT has been proposed in Ref. [20], which depicts the requirements for passive components such as energy storage capacitor (C_{bulk}), inductor (L), etc., as shown in Eqs. (3) and (4).

$$L \geq \frac{V_{DC(max)} t_{sw}}{k_{\Delta i} I_{L(min)}} \quad (3)$$

$$C_{bulk} \geq \frac{LI_{L(max)}^2}{2k_{\Delta v} V_{DC(min)}^2} \quad (4)$$

Here, $k_{\Delta i}$ and $k_{\Delta v}$ are respectively the current and voltage variation percentage, and t_{sw} is switching time. For a single device, L and C_{bulk} can be determined in a straightforward way once the ranges of V_{DC} and I_L are decided, and t_{sw} is obtained from the datasheet. For a generic design purpose, however, V_{DC} and I_L may fall out of the design, and t_{sw} is also very different among devices, which may require a modification or update on the designed DPT.

For the sake of compatibility, the design is to have the capacitor bank soldered on the PCB to minimize parasitics, while the inductor is connected to the PCB via screws and ring connectors, making it more flexible for reassembly. Our approach is fixing the value of C_{bulk} and only designing L . Once the inductance in use cannot meet the requirements of $k_{\Delta i}$ and $k_{\Delta v}$, a new inductor will be redesigned and adopted. In addition, considering the pre-stressing time in Eq. (2), which ought to be a positive value. Hence, based on Eqs. (1) – (4), a new design consideration for L is proposed as follows:

$$\min \left[\frac{2k_{\Delta V} C_{\text{bulk}} V_{DC(\min)}^2}{I_{L(\max)}^2}, \frac{t_{\text{stress}} V_{DC(\min)}}{I_{L(\max)}} \right] \geq L \geq \frac{V_{DC(\max)} t_{\text{sw}}}{k_{\Delta i} I_{L(\min)}} \quad (5)$$

To perform the characterization in a wide temperature range, a design trade-off between higher reliability and lower parasitics should be considered. For instance, if only the GaN devices are placed in the cryogenic environment, long gate wires and DC bus wires have to be applied since the rest of the DPT components are at room temperature. This will ensure that the power stage functions well, but will inevitably introduce severe parasitic inductance in the gate and power loop. On the other hand, if the overall platform is moved in the cryocooler, the performance of the power stage cannot be guaranteed due to the temperature impact on the value of L and C_{bulk} . In addition, the cooling capacity of the cryocooler should be capable enough for such a large thermal load.

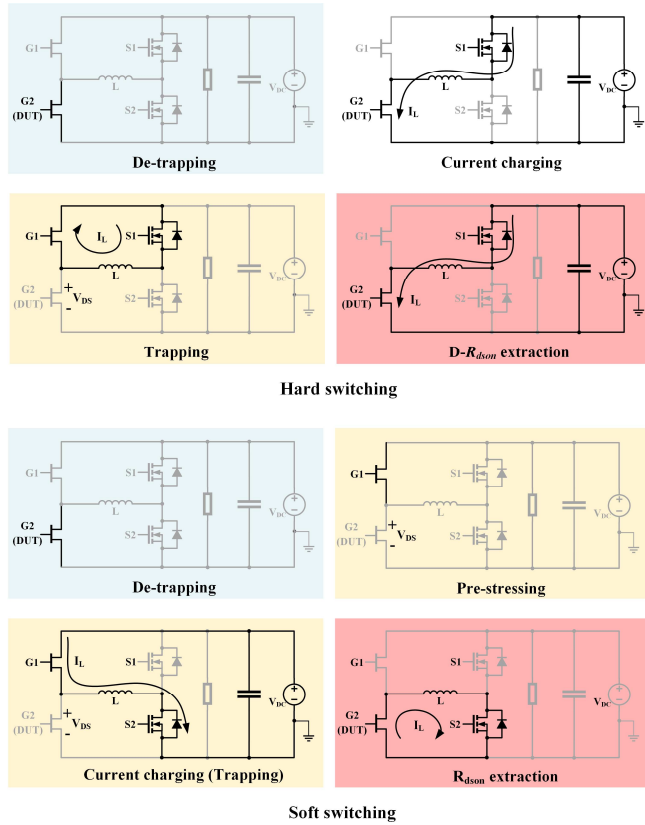


Fig. 5. Circuit operation in hard- and soft-testing modes.

In this paper, we separate the phase legs consisting of SiC and GaN into different PCBs. As shown in Fig. 3, the PCB that carries GaN devices, gate drivers and decoupling capacitors, is held inside the cryocooler, as the GaN board under test, while energy storage capacitors, inductor, SiC MOSFETs, etc. are at room temperature as the power stage, which will be detailed in the next subsection. Considering the wide temperature applied on the GaN board, the passive components should be carefully selected. It is suggested in Ref. [3] that NP0 capacitors and thin film resistors have good temperature consistency. However, the NP0 capacitors have relatively small capacitance compared to other types such as X7R. Since the energy storage capacitors are

placed at room temperature, therefore, NP0 capacitors will be only used for decoupling function which requires merely hundreds of nano-farad [20].

B. Electrothermal-mechanical Integration

To perform the cryogenic test down to 50 K, a LHe₂ cryocooler is utilized rather than LN₂. As shown in Fig. 6, dedicated design for the cryogenic test is conducted, including the vacuum enclosure, radiation shield and sample holder. The sample holder, which is made of copper, is mounted on the cryocooler's cooling station with indium as thermal interface. The radiation shield of polished aluminum is implemented on the cryocooler and surrounds the cooling station to block thermal radiation from the inner surface of vacuum enclosure. When performing the cryogenic test, a dry-type vacuum pump first operates to reduce the pressure to be ~ 10 mTorr, by which the heat convection is alleviated. Then, the compressor and cryocooler work to cool down the sample holder via the closed-loop cycling of LHe₂.

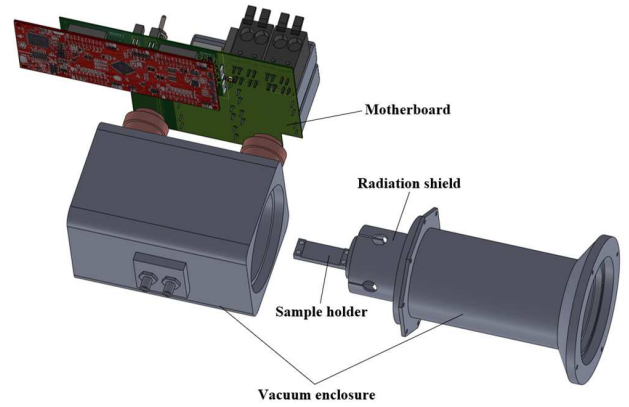


Fig. 6. Accessory design for the cryocooler.

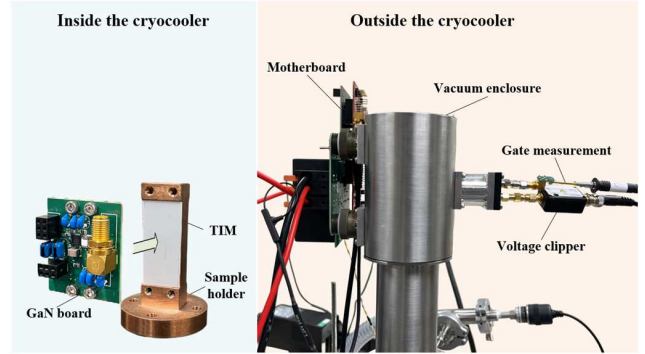


Fig. 7. Hardware implementation: DUT board, motherboard and assembly

Fig. 7 demonstrates the integration method of DPT hardware with the cryocooler setup. Inside the cryocooler, the GaN board is mounted on the sample holder made of copper, which actually acts as a cold head to cool down the device. Between the devices and sample holder, there is a thermal interface material to improve the cooling efficiency. Outside the cryocooler, to reduce the parasitics in the power loop, the motherboard carrying SiC MOSFETs, energy storage capacitors, etc. is directly fixed on the vacuum enclosure through two hermetic

feedthroughs, one of which is for the gate signals and the 12-V auxiliary DC voltage, and the other is for bus voltage. A hermetic dual SMA feedthrough is adopted for the measurement of V_{GS} and V_{DS} of G2, and a voltage clipper (CLP1500V15A1) is implemented to capture the small value of V_{DS} during the ON-state to thereby extract $D-R_{ds(on)}$.

C. Software Framework

Since multiple impacting factors are studied in this paper, the total number of tests N_{test} to be performed is determined as follows.

$$N_{\text{test}} = \prod n_{\sigma_i} \quad (6)$$

where n_{σ_i} is the number of testing points of the stressor σ_i . For instance, considering a testing plan that $V_{DC} = 40 \text{ V}, 60 \text{ V}, 80 \text{ V}$, $T = 50 \text{ K}, 90 \text{ K}, 120 \text{ K}, \dots, 400 \text{ K}$, $I_L = 19 \text{ A}, 28 \text{ A}, 38 \text{ A}, 48 \text{ A}$, $t_{\text{stress}} = 25 \mu\text{s}, 50 \mu\text{s}, 100 \mu\text{s}, 200 \mu\text{s}, 400 \mu\text{s}$, in both hard- and soft-switching modes, $N_{\text{test}} = 3 \times 14 \times 4 \times 5 \times 2 = 1680$, which is practically challenging to make it manually.

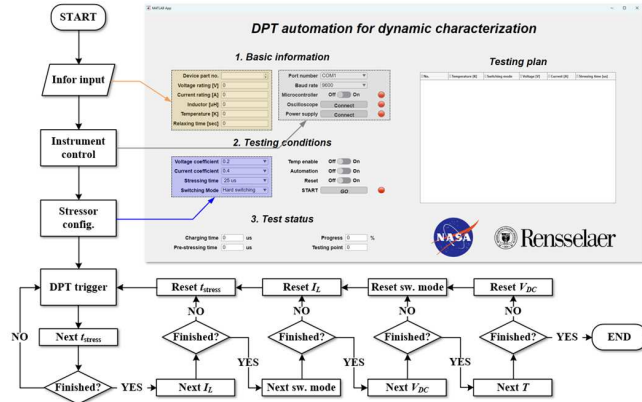


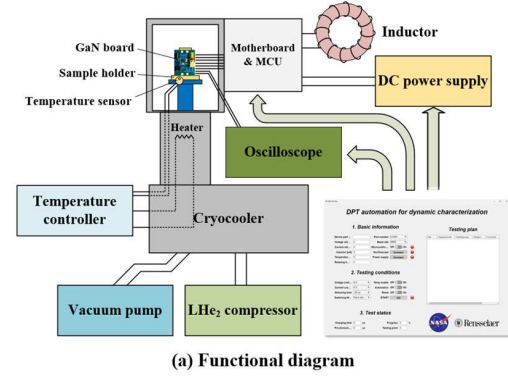
Fig. 8. Framework of the automated control.

To cope with this fact, a full automation function is essential to help save human effort and improve testing efficiency. As shown in Fig. 8, a graphical user interface (GUI) has been developed using MATLAB to realize automated configuration of stressors and measurement. The basic operation logic is: Firstly, the GUI establishes communication with the MCU, oscilloscope, DC power supply and temperature controller. Then, a table containing all required testing conditions is generated. The GUI will go through the table and convert stressors' value into commands for the MCU. For instance, if the current condition has $V_{DC} = 60 \text{ V}$ and $I_L = 28 \text{ A}$, the GUI will use Eq. (1) to calculate t_{charge} and convert it into the comparator level for ePWM, which can be directly executed by MCU. After that, the GUI will send out the commands to all controlled instruments, setting up voltage in the DC power supply, and making the MCU generate desired gate signals for S1, S2, G1 and G2. Simultaneously, the vertical and horizontal scale and position of the oscilloscope are adjusted corresponding to V_{DC} and I_L . Once the DPT begins, the oscilloscope will be triggered to capture waveforms, which will be named with a specific format and saved. By repeating such a procedure, the platform can operate 24/7 and thousands of datasets will be collected

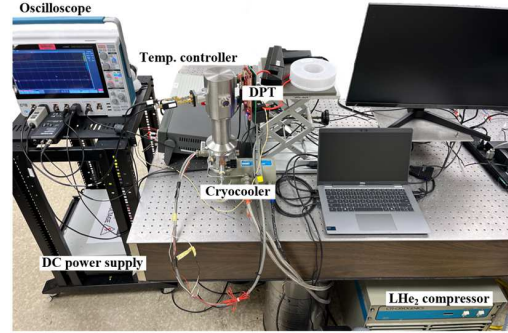
automatically. Roughly, performing the full test will take 18 – 20 hrs, 30% of which is for temperature setup.

IV. IMPLEMENTATION AND DATA COLLECTION

The implementation of the platform is illustrated in Fig. 9. To demonstrate the functionality of the platform, two commercial GaN e-mode HEMTs are used to build the phase leg for the GaN board. The DUT information and testing plan are given in TABLE II. The platform records waveforms of V_{ds} (through the voltage clipper) and I_L to obtain $D-R_{ds(on)}$ with $\frac{V_{ds}}{I_L}$. Simultaneously, V_{gs} is observed as well to check if the gate driver survives at cryogenic temperatures or not.



(a) Functional diagram



(b) Hardware implementation

Fig. 9. Implementation of the platform.

The rated gate driver output is 0 and +5 V, and the measured V_{gs} is presented in Fig. 10, showing that V_{gs} waveforms under different temperatures are almost overlapped. As temperatures vary from 50 K to 400 K, V_{gs} slightly changes between 4.97 V and 5.04 V, with a standard deviation of 0.023 V, indicating the temperature impact on DUT's gate voltage is negligible.

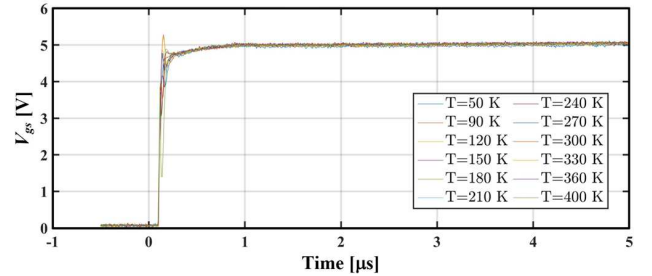


Fig. 10. V_{gs} at different temperatures (50 – 400 K).

TABLE II. DEVICE INFORMATION AND TESTING PLAN.

Package	Rated voltage [V]	Rated current [A]	Testing plan				
			T [K]	Sw. mode	V_{DC} [V]	I_L [A]	t_{stress} [μs]
QFN	200	48	50 – 400	Hard & soft	40 – 80	19 – 48	25 – 400

The raw data of $D-R_{dson}$ are presented in Fig. 11. Simply observing from the trend of the results, $D-R_{dson}$ under hard switching is overall higher than that at soft switching throughout the test, and is more sensitive to the applied stressors' value. This is revealed by the data dispersion, as $D-R_{dson}$ at hard switching varies from 8 – 22 m Ω , and at soft switching, $D-R_{dson}$ varies from 6 – 14 m Ω . The explanation is that due to the hot electron effect and higher switching losses at hard switching, which is related to $V_{ds} \times I_d$ at switching transient [13], [21], increasing testing voltages and currents enhances $D-R_{dson}$ more effectively than it does at soft switching, where the hot electron effect is largely eliminated.

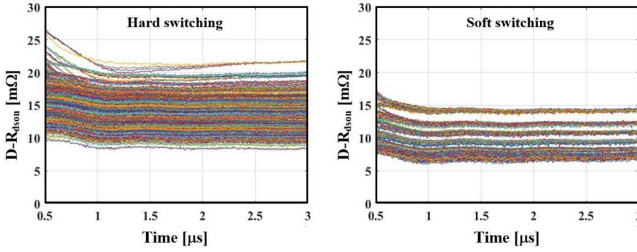


Fig. 11. Raw data of $D-R_{dson}$ at hard- and soft-switching modes, with various temperatures (50 – 400 K), voltages (40 – 80 V), currents (19 – 48 A) and stressing time (25 – 400 μs).

More quantitative comparison is made in Fig. 12, where $D-R_{dson}$ data are averaged and normalized by the measured static R_{dson} at each temperature point, with Eq. (7).

$$DR_{dson(norm)}(T) = \frac{1}{\Delta t} \int_{t_0}^{t_0+\Delta t} \frac{DR_{dson}(T, t)}{SR_{dson}(T)} dt \quad (7)$$

where t_0 is the time after turn-on to record $D-R_{dson}$ data ($t_0 = 0.5 \mu s$), Δt is the measurement window ($\Delta t = 2.5 \mu s$), $SR_{dson}(T)$ is the static R_{dson} as a function of temperature.

As shown in Fig. 12 (a), the static R_{dson} presents a non-monotonic variation with temperature increment: With temperature increasing, R_{dson} descends until around 180 K, and then keeps ascending. This is not common for GaN devices as they typically have the positive temperature coefficient of R_{dson} [3]. In Fig. 12 (b) – (d), normalized $D-R_{dson}$ versus temperature is visualized at hard- / soft-switching modes and compared in terms of varying V_{DC} , I_L and t_{stress} , respectively. Overall, the impact of $D-R_{dson}$ peaks at around 180 K, and at either higher or lower temperatures, $D-R_{dson}$ becomes less impactful. In terms of the influence of each stressor, it is found that temperatures and switching modes play the most significant role. The hard-switching $D-R_{dson}$ is around 2 – 4 times of the static value, and soft-switching one is 1-2 times of the static value, indicating the hard switching amplifies $D-R_{dson}$ effect almost two times as soft switching does, reflecting the contribution of hot electron effect as well as switching losses. Stressors including V_{DC} , I_L and

t_{stress} basically can enhance $D-R_{dson}$, but are way more pronounced at hard switching, while the results under soft switching are almost the same.

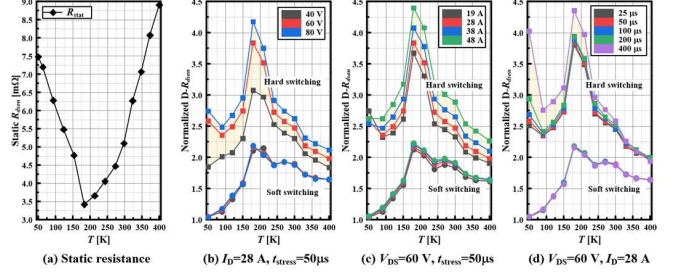


Fig. 12. $D-R_{dson}$ under different voltage, current and stressing time conditions.

The impacts of V_{DC} , I_L and t_{stress} are further quantified at cryogenic, room and high temperature conditions (respectively 50, 300 and 400 K), as shown in Fig. 13. In terms of hard-switching scenario, at room and high temperatures, separately increasing V_{DC} from 40 to 80 V and I_L from 19 to 48 A enhances $D-R_{dson}$ by around 30% and 38% of static R_{dson} , respectively, while increasing t_{stress} has no influence on $D-R_{dson}$. However, at 50 K, increasing I_L has less impact on $D-R_{dson}$ but increasing t_{stress} significantly contributes to $D-R_{dson}$.

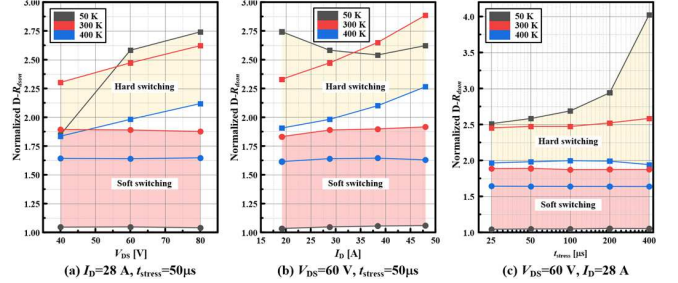


Fig. 13. Stressors' impacts on $D-R_{dson}$ at different temperatures (50 K, 300 K and 400 K).

Under soft switching, V_{DC} , I_L and t_{stress} almost imposes no impact on $D-R_{dson}$ at either cryogenic or high temperatures, indicating the electron trapping effect during the OFF state is negligible. This is possibly due to the applied blocking voltage is not sufficient to form a strong electric field between the drain and substrate and/or the drain and gate, since the maximum V_{DC} during this test is only 40% of DUT's voltage rating. With a hypothesis that the OFF state trapping is not sufficiently activated at low voltage level, the observation that t_{stress} does not influence $D-R_{dson}$ can be also explained. In this case, only hot electron effect contributes to $D-R_{dson}$.

V. CONCLUSIONS

In this paper, an automated platform for characterizing $D-R_{dson}$ of GaN HEMTs from 50 K to 400 K was developed. The platform features programmable stressors, including switching mode, voltage, current, and stress time, enabling comprehensive studies with high efficiency at which it automatically operates 24/7 and collects thousands of datasets within 20 hrs. Test results from a commercially available GaN HEMT reveal that $D-R_{dson}$ effect can vary non-monotonically with temperature, peaking at 180 K. Among the influencing factors, temperature and switching modes are found to be the most pronounced to impact $D-R_{dson}$. At hard switching, $D-R_{dson}$ is overall two times of that at soft switching, and it is more sensitively increased by the increment of voltages and current, especially at room and high temperatures. At soft switching, $D-R_{dson}$ is way less impacted by the stressors and almost remains unchanged, revealing that OFF-state electron trapping is not significant with the applied stressing conditions that cannot form a strong electric field within the device. In this case, hot electron effect is the dominant origin for $D-R_{dson}$ at hard switching. Therefore, soft switching is more preferable for low stressing applications for the sake of mitigating $D-R_{dson}$.

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